Mohamed (Seif) Mohamed

(Incoming) Tenure-Track Assistant Professor
Computer Science and Engineering Department, xx

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Research Areas

- Privacy-preserving & Communication-efficient Machine Learning
- Wireless Communications & Signal Processing
- Statistical Learning on Graphs & Network Data
- Generative AI & Trustworthy AI Systems

Employment & Research Experience

xx University — xx, USA
Director, NILE Research Lab — xx 2026 – Present

Princeton University — NJ, USA
Researcher, Wireless Systems Lab — Jun 2025 – present
Postdoctoral Research Associate — Jun 2023 – Jun 2025
Postdoctoral Research Fellow — Jun 2022 – Jun 2023
Mentors: Dr. Andrea J. Goldsmith and Dr. H. Vincent Poor

Nokia Bell Labs — NJ, USA
Research Collaborator — Oct 2024 – present
Visiting Researcher, Artificial Intelligence Research Lab — Sep 2024 - Oct 2024
Mentors: Dr. Iraj Saniee and Dr. Antti Koskela

The University of Arizona — AZ, USA
Graduate Research Assistant, Wireless Network and Cyber Security Research Lab — Jan 2017 - May 2022
Advisors: Dr. Ravi Tandon, and Dr. Ming Li

Nile University — Giza, Egypt
Graduate Research Assistant, Wireless Intelligent Networks Center — Oct 2014 - Dec 2016
Advisors: Dr. Karim Seddik, Dr. Amr El-keyi, and Dr. Mohammed Nafie

Egypt-Japan University of Science and Technology — Alexandria, Egypt
Undergraduate Research Intern, Wireless Research Center — Jun 2013 - Oct 2013; Jul 2014 - Aug 2014
Advisor: Dr. Moustafa Youssef

Alexandria University — Alexandria, Egypt
Undergraduate Research Assistant, Eelctrical Engineering Department — Oct 2013 – Jul 2014
Advisor: Dr. Essam Sourour

Education

Ph.D. **The University of Arizona**, Electrical and Computer Engineering — AZ, USA, May 2022
- Minor: Systems and Industrial Engineering
- Dissertation: Security and Privacy in Information Systems

M.Sc. **Nile University**, Communications and Information Technology — Giza, Egypt, Aug 2016
- Thesis: Interference Management in Spectrally and Energy Efficient Wireless Networks

B.Sc. **Alexandria University**, Electrical Engineering - Communications Section — Alexandria, Egypt, Jul 2014
- Graduation Project: Baseband Physical Layer Implementation of Software Defined Radio - WiFi 802.11 ac & LTE release 10 (Best Graduation Project)

Selected Publications

• Complete list of publications can be found here:

1. **Mohamed Seif**, Yasaman Ghasempour, H. Vincent Poor, Doru Calin, and Andrea Goldsmith, "Protecting Human Activity Signatures in Compressed IEEE 802.11 CSI Feedback", *IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2026
2. Hojjat Vavidan, **Mohamed Seif**, H. Vincent Poor, Ingrid Moreman, and Adnan Shahid, "Closed-loop Intelligence using Large Language Models in Wireless Networks,", *Proceedings of the 2025 IEEE/IFIP Wireless and Mobile Networking Conference (WMNC)*, 2025 *(Demo Paper)*
3. **Mohamed Seif**, Liyan Xie, Andrea Goldsmith, and Vincent Poor, "Differentially Private Online Community Detection for Cesnored Block Models: Algorithms and Fundamental Limits", *IEEE Transactions on Information Forensics & Security*, 2025
4. **Mohamed Seif**, Yuqi Nie, Andrea Goldsmith, and Vincent Poor, "Collaborative Inference over Wireless Channels with Feature Differential Privacy", *IEEE Journal of Selected Areas in Communications*, Special Issue on Intelligent Communications for Real-Time Computer Vision (Comm4CV), 2025
5. Joohyung Lee, **Mohamed Seif**, Jungchan Cho, and H. Vincent Poor, "Optimizing Privacy and Latency Tradeoffs in Split Federated Learning over Wireless Networks: A Joint Cut Layer Selection and Differential Privacy Approach", *IEEE Wireless Communication Letters*, 2024
6. Rajarshi Saha, **Mohamed Seif**, Andrea Goldsmith, and Vincent Poor, "Privacy Preserving Semi-Decentralized Mean Estimation over Intermittently-Connected Networks", *IEEE Transactions on Signal Processing*, 2024
7. **Mohamed Seif**, Yanxi Chen, Andrea Goldsmith, and Vincent Poor, "Differentially Private Sketch-and-Solve for Community Detection via Semidefinite Programming", *IEEE Journal of Selected Areas in Information Theory*, Special Issue on Information-Theoretic Methods for Trustworthy and Reliable Machine Learning, 2024
8. Bo Jiang, **Mohamed Seif**, Ravi Tandon and Ming Li, "Answering Count Query for Genomic Data under Perfect Privacy", *IEEE Transactions on Information Forensics & Security*, 2023
9. **Mohamed Seif**, Wei-Ting Chang, and Ravi Tandon, "Privacy Amplification for Federated Learning via User Sampling and Wireless Aggregation", *IEEE Journal of Selected Areas in Communications*, Special Issue on Distributed Learning over Wireless Edge Networks, 39 (12): 3821-3835, 2021.
10. Bo Jiang*, **Mohamed Seif***, Ravi Tandon, and Ming Li, "Context-Aware Local Information Privacy", *IEEE Transactions on Information Forensics & Security*, vol. 16: 3694-3708, 2021 **(*:co-first authors, alphabetical order)**
11. **Mohamed Seif**, Dung Nguyen, Anil Vullikanti, and Ravi Tandon, "Differentially Private Community Detection for Stochastic Block Models", *International Conference on Machine Learning (ICML)*, 2022 (Spotlight Presentation)
12. **Mohamed Seif**, Ravi Tandon, and Ming Li, "Wireless Federated Learning with Local Differential Privacy", *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, 2020 (Best Poster Award, Graduate Poster Symposium)

• Pending Papers (under review):

1. Liyan Xie, Muhammad Siddeek, **Mohamed Seif**, Andrea Goldsmith, and Mengdi Wang, "Detecting Post-generation Edits to Watermarked LLM Outputs via Combinatorial Watermarking", *Conference on Language Modeling (COLM)*, 2026
2. Hojjat Navidan, Mohammad Cheraghinia, Jaron Fontaine, **Mohamed Seif**, Eli De Poorter, H Vincent Poor, Ingrid Moerman, and Adnan Shahid, "Toward Autonomous O-RAN: A Multi-Scale Agentic AI Framework for Real-Time Network Control and Management", IEEE Network, 2026

• Patents:

1. Liyan Xie, **Mohamed Seif**, Mengdi Wang, and Andrea Goldsmith, "Watermarking and Post Edits Detection in Large Language Models", 2025 (filed)
2. **Mohamed Seif**, Andrea Goldsmith, and Vincent Poor, "Privacy-preserving AI over Next Generation Networks", U.S. Patent Application, No. 63/544,870, 2023 (filed)

Honors and Awards

- Invited Talk, 60th Annual **Allerton** Conference on Communication, Control, and Computing — Sep 2024
- Invited Talk, 2023 **INFORMS** Annual Meeting (🏆 Community Choice) — Oct 2023
- **Spotlight** Presentation, International Conference on Machine Learning **(ICML)** — July 2022
- Full Participation Grant Award, **ICML** — July 2022
- Best Poster Award, ECE Department, **The University of Arizona** — Mar 2020
- North American School of Information Theory Travel Grant, **Boston University** — Jul 2019
- Travel Grant Award, College of Engineering, **The University of Arizona** — May 2018
- Best Graduation Project Award, EE Department, **Alexandria University** — Aug 2014
- **Vodafone** Sponsorship for B.Sc. graduation project, **Alexandria University** — Aug 2013
- Academic Distinction Award, **Alexandria University** — 2011 – 2014

Recent Talks

- Invited Talk, Purdue Chip Design Center, **MediaTek, Inc.** — May 2026
- Talk, IEEE International Symposium on Spectrum Innovation **(DySPAN)**, Washington, D.C. — May 2026
- Invited Talk, **Nexcepta, Inc.** — May 2026
- Invited Talk, Computer Science and Engineering Department, **Oakland University** — April 2026
- Guest Lecture, Computer Science Department, **Georgia Southern University** — April 2026
- Talk, IEEE International Symposium on Information Theory **(ISIT)**, Ann Arbor, MI — Jun 2025
- Colloquium, Electrical Engineering and Computer Science Department, **Syracuse University** — Mar 2025
- Colloquium, Industrial and Systems Engineering Department, **Texas A & M University** — Feb 2025
- Colloquium, Electrical and Computer Engineering Department, **Johns Hopkins University** — Jan 2025
- Invited Talk, Electrical Engineering Department, **Stanford University** — Nov 2024
- Invited Talk, Wireless Systems Research Lab, **Intel** — Oct 2024
- Talk, **Asilomar** Conference on Signals, Systems, and Computers, Pacific Grove, CA — Oct 2024
- Invited Talk, Artificial Intelligence Research Lab, **Nokia Bell Labs** — Sep 2024
- Talk, University Center of Excellence (UCoE) Meeting, **U.S. Air Force Research Laboratory (AFRL)** — Aug 2024
- Invited Talk, Computer Science and Engineering Department, **University of North Texas** — Apr 2024
- Invited Talk, Computer Engineering Department, **San Jose State University** — Mar 2024
- Invited Talk, **INFORMS** Annual Meeting — Oct 2023

Teaching Experience

- ECE 539B: Special Topics in Information Sciences and Systems, **Princeton University** — Spring 2023
- ECE 696B - 310: Advanced Topics in Machine Learning and Applications, **The University of Arizona** — Spring 2021
- ECE 696B: Advanced Topics in Security, **The University of Arizona** — Fall 2020
- MENG 512: Signals and Systems, **Nile University** — Fall 2016

Grants & Funding

1. "Frequency-agile and -heterogeneous IoT networks", **BARI: Bilateral Academic Research Initiative** 2024
2. "Toward Privacy-preserving and Reliable Federated Learning for 6G Systems", ODFR: Innovation Fund for New Industrial Collaborations, **MediaTek** and **Princeton University** 2024
3. "Graph Mining and Network Science with Differential Privacy: Efficient Algorithms and Fundamental Limits", Secure and Trustworthy Cyberspace (SaTC), **U.S. National Science Foundation (NSF)** (**$** Awarded $1.2 Million) 2022
4. "Context Aware Privacy: Foundations, Algorithms and Applications", SaTC, **U.S. NSF** 2021

Mentorship and Advising

Ph.D. Students:

- Yanxi Chen, Princeton University (now at Alibaba Group)
- Ashlynn Crisp, Portland State University
- Zewei Deng, University of Minnesota
- Merkebu Tekaw Girmay, Ghent University–imec (now at OAI Foundation)
- Sherif Ghozzy, Princeton University
- Atsutse Kludze, Princeton University
- Mostafa Naseri, Ghent University–imec
- Hojjat Navidan, Ghent University–imec
- Yuqi Nie, Princeton University (now at NovaQuant)
- Rajarshi Saha, Stanford University (now at Amazon Annapurna Labs)
- Faranak Solat, Gachon University (now at Sejong University)

Service

Guest Editor:
- Entropy, Special Issue on Information-Theoretic Security and Privacy 2024–2026

Session Chair:
- IEEE CISS 2024

TPC Member:
- IEEE ICC 2026
- IEEE ICNC 2025
- IEEE GLOBECOM 2022–2026

Technical Reviewer:
- Conferences: NeurIPS, ICML, AISTATS, GLOBECOM, ISIT, INFOCOM 2015–present
- Journals: ComMag, JSAC, JSAIT, T-IFS, T-IT 2017–present

Technical Skills

Programming: Python, MATLAB

Machine Learning & AI: PyTorch, scikit-learn, NumPy, SciPy, LLMs, Generative AI, Federated Learning, Edge AI

Privacy & Optimization: Differential Privacy, Gurobi, CVX

Wireless & Signal Processing: MIMO, OFDM, 5G NR fundamentals, Wi-Fi (802.11ac/ax), CSI feedback, beamforming